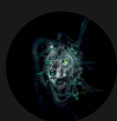
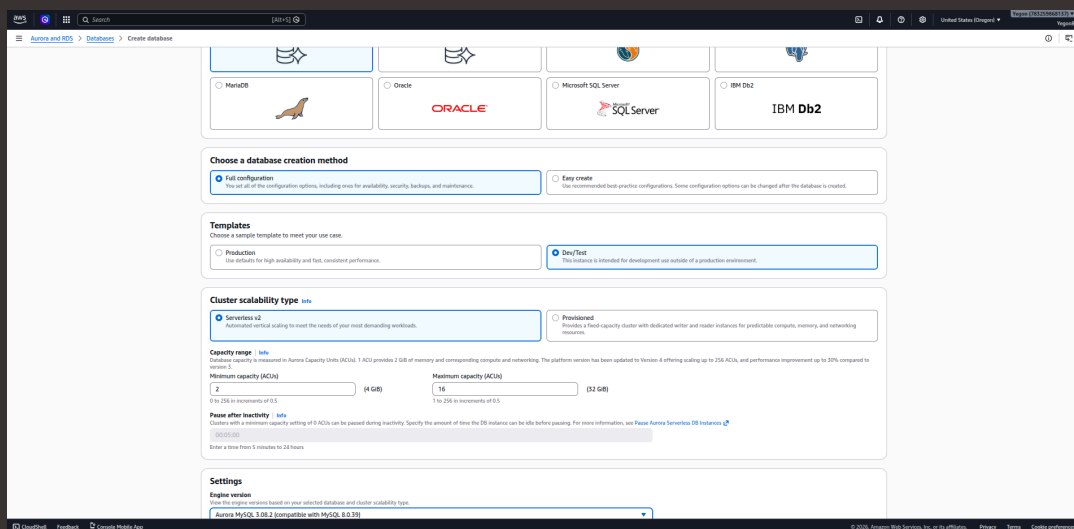


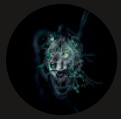


Connect a Web App to Amazon Aurora



Kiprono Yegon





Introducing Today's Project!

What is Amazon Aurora?

Amazon Aurora is a fully managed relational database service compatible with MySQL and PostgreSQL that combines the performance and availability of high-end commercial databases with the simplicity and cost efficiency of open-source databases. It is useful because it provides automatic backups, high availability, fault tolerance, fast failover, scalable storage, and read replicas for handling large workloads, while reducing the operational overhead of managing database infrastructure manually.

How I used Amazon Aurora in this project

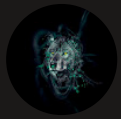
In today's project, I used Amazon Aurora to create a database that will be used in the oncoming projects and also connected it to an EC2 instance

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the ease of connecting the Amazon Aurora to the EC2 instance

This project took me...

The project took me about 45 minutes

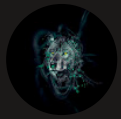


In the first part of my project...

Creating an Aurora Cluster

A relational database is a type of database that organizes data into tables, which are collections of rows and columns., the rows relate to the columns and vice-versa

Aurora is a good choice when in need for something large-scale, with peak performance and uptime. This is because Aurora databases use clusters. Ordinary relational databases, like MySQL and Oracle are more generic and cost-effective. They suit smaller databases and less demanding workloads.



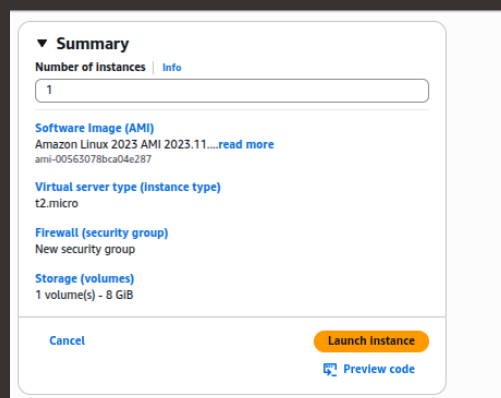
Halfway through I stopped!

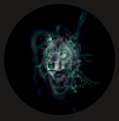
I stopped creating my Aurora database because there was no existing running instance. Create the instance so that it connect to the aurora database.

Features of my EC2 instance

I created a new key pair for my EC2 instance because the key pair in EC2 is quite literally the keys to access our EC2 instance. We need keys to our EC2 instance if we want to add, change, or update how our EC2 instance is running.

When I created my EC2 instance, I took particular note of the Public IPv4 DNS and the Key pair name





Then I could finish setting up my database

The screenshot shows the 'Connectivity' section of the AWS Aurora Database console. It includes options for connecting to an EC2 compute resource, selecting an EC2 instance, and choosing a VPC. The 'Connect to an EC2 compute resource' option is selected. The EC2 instance 'i-0ae3edcd985282d51' is chosen. The VPC 'Default VPC' is selected. A warning message states: 'Some VPC settings can't be changed when a compute resource is added. Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.'

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

EC2 instance [Info](#)
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0ae3edcd985282d51
nextwork-ec2-instance-web-server

Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Network type [Info](#)
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.

Default VPC (vpc-058fca61c8122ebc)
4 Subnets, 4 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

Aurora Database uses clusters because each cluster consists of a primary instance (where all write operations occur) and multiple read replicas as back-ups. If your database's primary instance fails, one of the replicas can be promoted to primary automatically.



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